PRACTICE SHEET

1. The value of $2^{2-\log_2 5}$ is equal to:
(a) 4/5 (b) 5/4
(c) 2/5 (d) 5/2

2. If $y = 2^{1/\log_x(8)}$, then x is equal to:
(a) y (b) y^2
(c) y^3 (d) None of these

3. If a, b and c are in GP, then $\log_{ax} x$, $\log_{bx} x$, $\log_{cx} x$ are in:
(a) GP (b) HP
(c) AP (d) None of these

4. If $2\log(x+1) - \log(x^2-1) = \log 2$, then x equals to:
(a) 1 (b) 0
(c) 2 (d) 3

5. The number of solutions of $\log_2(x-1) = 2\log_2(x-3)$:
(a) 2 (b) 1
(c) 6 (d) 7

6. The value of $e^{(\log_{10}\tan 1^\circ + \log_{10}\tan 2^\circ + \ldots\ldots + \log_{10}\tan 89^\circ)}$ is equal to:
(a) 0 (b) 1
(c) e (d) 1/e

7. How many numbers of digits are there in 2^{98}? (Given that $\log_{10} 2 = 0.30103$)
(a) 98 (b) 99
(c) 30 (d) 29

(a) – 0.0512 (b) 0.4213
(c) 0.3010 (d) None of these

15. The value of $3^{2-\log_3(4\log 36)}$ is:
(Given log 2 = 0.3010, log 3 = 0.4771)
(a) 1.45 (b) 0.69
(c) 1.2 (d) 3.2

16. If $\log_{10}\sqrt{1+x} + 3\log_{10}\sqrt{1-x} = \log_{10}\sqrt{1-x^2} + 2$, then the value of x is:
(a) 4 (b) –99
(c) 5 (d) None of these

17. $\log_{25} 625 - \log_{31} 961 + \log_{29} 841 =$
(a) 2 (b) 6
(c) 0 (d) 4

18. If $\log x + \log y = \log(x+y)$ then :
(a) x = y (b) xy = 1
(c) $y = \frac{x-1}{x}$ (d) $y = \frac{x}{x-1}$

19. If a , b , c are in GP then $\log_a n$, $\log_b n$, $\log_c n$ are in
(a) A.P. (b) G.P.
(c) H.P. (d) not

20. x triples every second . How will $\log_2 x$ change every